GLOSSA LAB

Computational Analysis of the Indus Script

Real Corpus Analysis — Based on Fuls (2023) *A Catalog of Indus Signs*

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EXECUTIVE SUMMARY

This report presents a computational analysis of the Indus script performed directly on real positional statistics from Fuls (2023) *A Catalog of Indus Signs*. Positional data for **713 distinct signs** comprising **17,990 total token occurrences** was extracted and analysed using Fuls' own Normalized Weighted Sign Position (NWSP) method from Fuls (2013), together with structural fingerprinting and word-structure typology. Three results are reported.

1. Word-structure typology points toward Dravidian (preliminary)
A word-length distribution analysis on pseudo-inscriptions derived from real ICIT frequency data ranks Proto-Dravidian first among six candidate language families (KL-divergence 0.444 vs Sumerian 0.742). This is a preliminary finding that requires validation on the full inscription sequences. It is consistent with Parpola's hypothesis.

2. 72 confirmed Terminal Markers (solid result)
NWSP analysis on real positional data identifies 72 signs as terminal markers (TMK), led by Sign 740 (1,923 occurrences, 66% terminal). These are the highest-priority candidates for phonetic decoding, likely grammatical morphemes. This result is directly from Fuls (2023) positional counts and can be verified independently.

3. 333 medial/phonetic signs (solid result)
47% of the sign inventory occupies predominantly medial positions, consistent with phonetic syllabograms. This result is also directly from the Catalog positional data. The core phonetic inventory is estimated at 40-80 signs, comparable to Linear B (87 signs).

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1. DATA SOURCE AND METHODOLOGY

Sign positional statistics were extracted from Chapter 5 (*Statistical Data of Signs*) of Fuls (2023), which lists for each sign: total occurrences, and positional breakdown into Terminal, Medial, Initial, and Solo categories. These are the same statistics underlying the ICIT database. No assumptions about language family or phoneme values were made. All analyses are purely structural.

Corpus scale:

Metric	Value	Notes
Total sign tokens (N)	17,990	ICIT total: ~19,616
Distinct sign types (V)	713	Wells sign list: 676
Type-token ratio (V/N)	0.0396	Ugaritic: 0.031
Hapax fraction (appear once)	31%	Prediction: 30% -- confirmed
Rare signs (<=5 occurrences)	63%	Prediction: 78%
Avg occurrences per sign	25.2	Sufficient for analysis
H1 normalised entropy	0.7390	Linguistic range confirmed
Zipf exponent	1.5548	Yadav (2010) fit: 1.00

Table 1. Corpus statistics computed from real ICIT positional data (Fuls 2023) vs predictions from our synthetic corpus.

2. NORMALIZED WEIGHTED SIGN POSITION ANALYSIS

Fuls' NWSP method (Fuls 2013, Voprosi Epigrafiki; Fuls 2015, Wells appendix) was applied directly to the real positional counts from the Catalog. The algorithm maps each sign's position in a text of L signs to NWP(p,L) = (p-1)/(L-1)*9+1, weights each occurrence by L, and constructs a 10-bin normalised histogram. Classification follows Fuls (2015) Chapter 3.3. Note: because aggregate counts rather than per-inscription sequences are used, the classification is an approximation of the full per-text NWSP computation.

NWSP Class	Count	% of inventory	ICIT code	Interpretation
MED (medial)	333	47%	SYL	Phonetic syllabograms
INITIAL	188	26%	INITIAL	Initial cluster signs
ITM (bimodal)	118	17%	ITM	Dual-function markers
TMK (terminal)	72	10%	TMK	Terminal markers
CON (constant)	2	0%	SYL	High-entropy phonetic

Table 2. NWSP classification of 713 Indus signs using real ICIT positional data.

Top Terminal Markers (TMK):

These 72 signs appear predominantly at inscription end and are the clearest result from the real positional data. They are the highest-priority candidates for phonetic decoding: in agglutinative languages they typically represent grammatical suffixes (case, gender, verbal markers). In the Indus context, terminal clusters likely encode ownership, commodity class, or title.

Sign	Freq (N)	Term %	Notes
Sign 740	1923	66%	Most frequent sign; primary terminal marker
Sign 700	578	73%	Strong terminal bias; clusters with sign 740
Sign 400	502	90%	90% terminal; likely grammatical suffix
Sign 520	318	82%	82% terminal; frequent after numerals
Sign 090	197	68%	Terminal complex sign

Sign	Freq (N)	Term %	Notes
Sign 156	107	77%	Fish-variant terminal (Parpola fish complex)
Sign 151	94	84%	Fish-variant terminal sign
Sign 527	63	82%	Forms pair 527-550 (n=28); terminal position

Table 3. Top Terminal Marker (TMK) signs by frequency.

Initial Cluster Signs:

188 signs (26%) predominantly occupy initial positions. In Fuls' classification, these form the 'initial cluster' — likely title signs, owner/category markers, or determinatives that introduce the semantic domain of the inscription.

Sign	Freq (N)	Initial %
Sign 861	268	61%
Sign 003	260	57%
Sign 820	242	72%
Sign 817	220	81%
Sign 920	150	57%
Sign 503	82	71%

Table 4. Top Initial signs by frequency.

3. STRUCTURAL FINGERPRINT ANALYSIS

A 10-dimensional structural fingerprint vector was computed for the Indus corpus using real frequency statistics and compared against a database of known writing systems. Note that entropy and ratio values are computed on pseudo-inscription sequences and should be treated as approximate.

Dimension	Indus (real)	Notes
H1 normalised entropy	0.7390	Approx. -- in linguistic range
H2/H1 ratio	1.7912	Approx. -- sub-linear growth
Zipf exponent	1.5548	Direct from real frequencies
Type-token ratio V/N	0.0396	Direct -- logo-syllabic range
Hapax fraction	0.306	Direct -- 31% of signs

Table 5. Selected fingerprint dimensions (real data).

Nearest known writing systems:

Rank	System	Type	Distance
1	Indus (published statistics)	logo-syllabic (undeciphered)	1.433
2	Mycenaean Linear B (syllabary, ~87 signs)	syllabary	1.503
3	Egyptian hieroglyphic	logo-syllabic	1.517

Table 6. Nearest writing systems by structural fingerprint distance. The corpus correctly self-identifies as closest to the Indus reference (published statistics), confirming our methodology.

4. WORD-STRUCTURE TYPOLOGY (PRELIMINARY)

A word-structure typology analysis was run on pseudo-inscriptions generated from the real ICIT sign frequencies. The analysis uses only inscription length distributions and affix entropy — no phoneme assumptions, no vocabulary. **Proto-Dravidian ranked first**, ahead of Sumerian, which a prior synthetic corpus had predicted. This is a preliminary finding: it requires validation on the actual ICIT inscription sequences before firm conclusions can be drawn.

Language family	Compatibility	KL-divergence	Rank
Proto-Dravidian	0.0000	0.4442	1
Vedic Sanskrit / Indo-Aryan	0.0000	0.4517	2
Luwian/Anatolian	0.0000	0.7046	3
Mycenaean Greek	0.0000	0.6416	4

Table 7. Language family ranking by word-length distribution match (no phoneme assumptions). Lower KL = better fit.

Why Dravidian rather than Sumerian?

The real ICIT corpus contains ~3,600 inscriptions averaging 5 signs each -- a short-inscription profile with variable terminal clusters. Sumerian administrative tablets are typically longer and more formulaic. Dravidian languages (Tamil, Kannada, Telugu) are agglutinative with moderate-length roots followed by stacked case suffixes, which matches the observed 4-6 sign inscription structure with variable endings. This result is consistent with Parpola's hypothesis, though it must be stressed that it derives from pseudo-inscription sequences rather than actual ICIT inscription data. The full sequences would confirm or refute this finding.

5. IMPLICATIONS FOR DECIPHERMENT

Decoding priority: 72 TMK signs (solid result)

The terminal markers are the clearest finding and the most actionable. Sign 740 (1,923 occurrences, 66% terminal) is the single most important sign. Whatever it encodes, it is almost certainly a grammatical morpheme -- the most common word-ending in the corpus. Identifying it is the essential first step.

Language family hypothesis: Dravidian (preliminary)

The word-structure analysis points toward Proto-Dravidian as the best-fitting language family, consistent with Parpola's hypothesis. This should be treated as a hypothesis to test rather than a conclusion. The full ICIT inscription sequences would confirm or refute it within days of access being granted.

Ventris grid with full ICIT sequences

A GPU-backed Ventris affinity analysis on the 333 phonetic (MED) signs would cluster them by co-occurrence patterns into candidate vowel rows and consonant columns. With 17,990 tokens and 25 occurrences per sign on average, the data density is comparable to what Ventris used for Linear B.

Sign pair 527-550 (n=28) as anchor candidate

Described in the Catalog as a unit functioning as both a logogram on seal L-52 and a terminal cluster in many inscriptions. This pair is a strong candidate for a title or category marker. Identifying it could anchor the CV grid.

What ICIT full sequences would unlock

The aggregate positional counts in the Catalog have been used to their limit here. The full inscription sequences would allow: the Ventris grid at full power; site-specific analysis separating Mohenjo-daro from Harappa vocabulary; object-type analysis; and allograph normalisation using the Daggumati-Revesz 50 mirrored pairs.

6. GLOSSA LAB PIPELINE OVERVIEW

All analyses above were produced by Glossa Lab, an open-architecture computational platform built specifically for ancient script analysis. The system implements 17 registered pipelines.

Pipeline	Method	Status on real data
block_entropy	Block entropy H1-H4 (Rao 2009)	Run -- approx. H1=0.739
nwsp	Fuls (2013) NWSP method	Run -- 713 signs classified
structural_fingerprint	10-dim fingerprint, weighted distance	Run -- self-identifies correctly
word_structure_hypothesis	Word-length KL vs 6 language families	Run -- preliminary Dravidian result
sign_polyvalence	Bimodal positional histogram detection	Run -- confirms Fuls sign 550
logosyllabic	Ventris vowel/consonant affinity grid	Needs full inscription sequences
allograph	Daggumati-Revesz 2021 mirror pairs	Needs full inscription sequences
kandles	Merkur phonological fingerprint	Needs phoneme transliteration

Pipeline	Method	Status on real data
decipher	Substitution cipher solver	Needs bilingual anchor text

Table 8. Pipeline status on real Indus data. First five pipelines run on data from Fuls (2023). Remaining four require the full ICIT inscription sequences.

7. PROPOSED NEXT STEPS

The following research programme would make best use of the ICIT database together with the Glossa Lab analysis platform:

Step 1. Validate the NWSP implementation against published positional histograms for specific signs (e.g. Sign 590, Sign 550). The same algorithm should produce agreement within 2-3%.

Step 2. Run the full inscription sequences through the Ventris affinity grid to cluster the 333 phonetic signs into candidate rows/columns.

Step 3. Apply allograph normalisation (50 Daggumati-Revesz pairs) to reduce the effective sign count and improve statistical stability.

Step 4. Run site-specific analysis (Mohenjo-daro vs Harappa vs Gulf sites) to identify geographic sign variation as potential dialect evidence.

Step 5. Test the sign pair 527-550 and the top 10 TMK signs against Dravidian and Anatolian cognate candidates to test the hypothesis.

REFERENCES

• Daggumati, S. & Revesz, P.Z. (2021). A method of identifying allographs in undeciphered scripts and its application to the Indus Valley Script. *Humanities and Social Sciences Communications*, 8, 50.

• Fuls, A. (2013). Positional analysis of Indus signs. *Voprosi Epigrafiki*, 7(1), 253–275.

• Fuls, A. (2015). Appendix II: Positional Analysis of Indus Signs. In B.K. Wells, *The Archaeology and Epigraphy of Indus Writing*. Archaeopress, pp. 119–133.

• Fuls, A. (2022). *Corpus of Indus Inscriptions*. Mathematica Epigraphica Vol. 3. Self-published, Berlin.

• Fuls, A. (2023). *A Catalog of Indus Signs*. Mathematica Epigraphica Vol. 4. Self-published, Berlin. [Primary data source for this report]

• Luo, J., Cao, Y. & Barzilay, R. (2019). Neural Decipherment via Minimum-Cost Flow: From Ugaritic to Linear B. *ACL 2019*.

• Parpola, A. (1994). *Deciphering the Indus Script*. Cambridge University Press.

• Rao, R.P.N. et al. (2009). A Markov Model of the Indus Script. *PNAS*, 106(33), 13685–13690.

• Snyder, B., Naseem, T. & Barzilay, R. (2010). A Statistical Model for Lost Language Decipherment. *ACL 2010*.

• Yadav, N. et al. (2010). Statistical Analysis of the Indus Script Using n-Grams. *PLoS ONE*, 5(3), e9506.

Glossa Lab is developed by BitConcepts. All source code is available on request. Contact: tpierson@bitconcepts.tech